

IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

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DECLARATION

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1 INTRODUCTION

1.1 Background Study

Aquaculture has become one of the fastest-growing food production sectors in the world, contributing significantly to global food security and economic development. In small and medium-scale fish farms, daily operations depend heavily on continuous monitoring and resource management, especially electricity usage, feed consumption, and water quality control. Equipment such as aerators, pumps, and filtration systems operate for long durations, often leading to high energy consumption and increased operational costs.

Despite the importance of cost control, many aquaculture farms still rely on traditional management approaches such as manual record keeping, monthly utility bills, and basic spreadsheets. These methods provide only a limited view of expenses and fail to capture real-time variations in cost drivers. As a result, farmers face difficulties in identifying inefficient practices, comparing production cycles, and planning budgets effectively.

Recent advancements in Internet of Things (IoT) technologies have introduced new opportunities for improving farm management. IoT-based systems enable continuous monitoring of environmental and operational parameters using sensors and embedded devices. These systems can collect, process, and transmit data in real time, allowing farmers to make timely decisions. However, most existing IoT solutions in aquaculture focus mainly on water quality monitoring and fish health, while financial management and cost analytics remain underdeveloped.

In this context, integrating energy monitoring, expense tracking, and predictive analytics into a unified system becomes essential. Such an approach can provide a comprehensive understanding of farm operations by linking resource usage with financial outcomes. This study focuses on developing a cost analytics and forecasting module within an IoT-enabled aquaculture system to address these challenges.

1.2 Problem Statement

Small-scale aquaculture farms often lack an effective and affordable system to monitor and manage their operational costs in real time. Electricity consumption, which is a major expense, is typically measured using monthly bills that do not provide detailed insights into usage patterns. At the same time, other expenses such as feed, labour, chemicals, and maintenance are recorded manually, leading to inconsistencies and delays.

This fragmented approach creates several problems. Farmers are unable to accurately calculate key performance indicators such as cost per fish or total production cost per cycle. They also struggle to detect abnormal cost increases or inefficiencies in resource usage. Furthermore, the absence of predictive tools prevents proactive budgeting and planning for future production cycles.

Although IoT technologies have been applied in aquaculture, existing systems rarely integrate energy monitoring with structured expense logging and financial analytics. As a result, there is no unified platform that connects real-time operational data with cost analysis and forecasting.

Therefore, the main problem addressed in this research is the lack of a low-cost, integrated system that can combine energy monitoring, expense tracking, cost analytics, and predictive forecasting for small-scale aquaculture farms.

1.3 Research Gap

A review of existing literature shows that most aquaculture IoT systems are designed to monitor environmental parameters such as temperature, pH, dissolved oxygen, and turbidity. While these systems are useful for maintaining fish health, they do not address financial

management aspects. Studies on energy monitoring in other domains, such as smart homes and industrial systems, demonstrate the effectiveness of using sensors to track electricity consumption and reduce energy waste [1].

Similarly, forecasting techniques such as ARIMA and exponential smoothing have been widely used in energy and cost prediction in sectors like buildings and power systems [2]. However, these approaches are rarely applied in aquaculture, particularly in small-scale farming environments. Moreover, existing financial tools and POS systems are not integrated with IoT-based data collection, limiting their ability to provide real-time insights.

Another key limitation is the lack of cycle-based analysis. Aquaculture production operates in cycles, and comparing costs across cycles is essential for improving efficiency. Existing systems do not effectively combine telemetry data, expense records, and stock information to generate meaningful KPIs such as cost per fish or cost per kilogram.

This research addresses these gaps by proposing an integrated module that combines IoT-based energy monitoring with POS-based expense tracking and applies predictive analytics to generate actionable insights.

1.4 Research Objectives

1.4.1 Main Objective

To design and develop a low-cost IoT-based cost analytics and forecasting system for aquaculture that integrates energy monitoring, expense tracking, and predictive modeling to support efficient financial management.

1.4.2 Specific Objectives

- To implement site-level energy monitoring using sensors and an ESP32 microcontroller.
- To develop a structured POS module for recording farm expenses.
- To calculate key financial indicators such as total cost, cost per fish, and category-wise cost distribution.
- To apply time-series forecasting techniques to predict future energy consumption and operational costs.
- To provide real-time alerts for abnormal conditions such as energy spikes and budget deviations.
- To enable data visualization and reporting through dashboards and exportable reports.

1.5 Scope of the Study

This study focuses on the development of a cost analytics and forecasting module within an IoT-based aquaculture system. The system is designed for small and medium-scale fish farms and emphasizes affordability, ease of deployment, and practical usability.

The scope includes energy monitoring at the main power supply, manual entry of operational expenses through a POS interface, and the generation of financial analytics and forecasts. The system also incorporates alert mechanisms and reporting features to support decision-making. However, the study does not cover large-scale industrial aquaculture systems or advanced machine learning models requiring extensive datasets. The implementation is limited to a prototype-level system suitable for real-world testing in small farm environments.

1.6 Significance of the Study

This research contributes to the field of smart aquaculture by addressing an important but often overlooked aspect of farm management, which is cost optimization. By integrating IoT technology with financial analytics, the proposed system provides a comprehensive solution for monitoring and managing operational expenses.

The system enables farmers to gain better visibility into their cost structure, identify inefficiencies, and make informed decisions. The inclusion of predictive forecasting allows

proactive planning, reducing financial risks and improving overall productivity. Additionally, the low-cost design ensures that the solution is accessible to small-scale farmers, promoting wider adoption of digital technologies in aquaculture.

Great I'll continue with a **strong, detailed, human-written Chapter 2** in IEEE style with proper in-text citations like [1], [2].

2 LITERATURE REVIEW

2.1 Introduction

The integration of technology into aquaculture has gained significant attention in recent years due to the increasing demand for efficient and sustainable fish farming practices. Researchers have explored various approaches to improve monitoring, automation, and decision-making in aquaculture systems. These approaches include Internet of Things (IoT) based monitoring, smart energy management, cost analysis systems, and predictive modeling techniques.

This chapter reviews existing studies and technologies related to aquaculture monitoring, energy consumption analysis, cost management, and forecasting. It also identifies the limitations of current systems and highlights the need for an integrated solution that combines these aspects into a single platform.

2.2 IoT in Aquaculture Systems

IoT technology has transformed traditional aquaculture practices by enabling real-time monitoring and automation. IoT-based systems use sensors, microcontrollers, and communication networks to collect and transmit data related to environmental and operational conditions.

Several studies have focused on monitoring water quality parameters such as temperature, pH, dissolved oxygen, and turbidity. For example, Saha et al. proposed an IoT-based system that continuously monitors water conditions and sends alerts when parameters exceed safe limits [3]. These systems help farmers maintain optimal conditions for fish growth and reduce mortality rates.

In addition to environmental monitoring, IoT devices allow remote access and control of farm equipment. Farmers can monitor their ponds using mobile or web applications, reducing the need for constant physical presence. Despite these advantages, most IoT-based aquaculture systems primarily focus on fish health and environmental conditions, with limited attention given to financial aspects such as cost tracking and budgeting.

2.3 Energy Consumption in Aquaculture

Energy consumption is a major operational cost in aquaculture systems. Equipment such as aerators, pumps, and filtration units require continuous electricity supply, leading to significant energy expenses.

Research conducted by Badiola et al. highlights that energy usage varies depending on system design, species, and environmental conditions [1]. The study emphasizes the importance of monitoring energy consumption to improve efficiency and reduce costs. Similarly, Boyd and McNevin analyzed energy use in shrimp farming and identified aeration as one of the largest contributors to electricity consumption [2].

Traditional energy monitoring methods rely on monthly utility bills, which provide only aggregated data. This makes it difficult to identify specific periods of high consumption or inefficient equipment usage. IoT-based energy monitoring systems address this limitation by providing real-time data on power usage.

Smart energy monitoring systems using sensors such as ACS712 and voltage transformers have been widely used in smart homes and industrial environments. These systems measure current and voltage to calculate power and energy consumption. However, such solutions are rarely adapted specifically for aquaculture environments, and they often lack integration with financial data.

2.4 Cost Management and Expense Tracking Systems

Effective cost management is essential for the sustainability of aquaculture operations. Traditional methods involve manual record keeping using notebooks or spreadsheets. These methods are prone to errors, inconsistencies, and delays, which reduce their reliability.

Point-of-Sale (POS) systems are commonly used in business environments to record transactions and manage expenses. In aquaculture, POS systems can be adapted to track costs such as feed, labour, chemicals, and maintenance. However, existing POS systems are not designed specifically for aquaculture and do not integrate with IoT-based monitoring systems. Studies on farm management systems indicate that integrating expense tracking with operational data can improve decision-making. However, most existing systems treat financial data and operational data separately. This lack of integration prevents farmers from understanding the relationship between resource usage and costs.

2.5 Predictive Forecasting Techniques

Forecasting plays a crucial role in planning and decision-making. In aquaculture, forecasting can be used to predict future energy consumption, operational costs, and production outcomes. Time-series forecasting methods such as ARIMA (AutoRegressive Integrated Moving Average) and Exponential Smoothing (ETS) are widely used in various domains. Hyndman and Athanasopoulos demonstrated that these models are effective for short-term forecasting, especially when dealing with limited datasets [4].

In energy management, forecasting models help predict electricity demand and optimize usage. Similarly, in financial systems, forecasting is used to estimate future expenses and revenues. However, the application of these techniques in aquaculture cost management is still limited.

Most existing forecasting studies assume the availability of clean and structured datasets. In small-scale aquaculture farms, data is often incomplete or inconsistent, making it challenging to apply advanced models. Therefore, there is a need for simple and efficient forecasting techniques that can work with limited data.

2.6 Smart IoT Communication and Cloud Integration

Modern IoT systems rely on communication protocols and cloud platforms for data transmission, storage, and processing. MQTT (Message Queuing Telemetry Transport) is a lightweight protocol widely used in IoT applications due to its efficiency and low bandwidth requirements.

Cloud platforms such as Firebase and Supabase provide real-time databases, authentication, and API services. These platforms enable seamless data synchronization and remote access to system data.

Research shows that combining IoT devices with cloud services enhances system scalability and reliability. Real-time dashboards and mobile applications allow users to monitor system performance and receive alerts instantly. However, many existing implementations focus only on data visualization and do not incorporate advanced analytics or forecasting features.

2.7 Limitations of Existing Systems

Based on the literature review, several limitations can be identified in existing aquaculture and IoT-based systems:

- Most systems focus only on environmental monitoring and ignore financial management.
- Energy monitoring solutions are not tailored for aquaculture and lack integration with expense data.
- Traditional cost tracking methods are manual and prone to errors.
- Forecasting techniques are not widely applied in aquaculture cost analysis.
- Existing systems do not provide cycle-based cost analysis or key performance indicators such as cost per fish.
- Industrial solutions are often expensive and not suitable for small-scale farms.

These limitations highlight the need for a unified system that integrates monitoring, cost tracking, analytics, and forecasting.

2.8 Summary

This chapter reviewed the existing research related to IoT-based aquaculture systems, energy monitoring, cost management, and forecasting techniques. While significant progress has been made in environmental monitoring and automation, there is still a gap in integrating financial analytics with operational data.

The proposed system addresses these gaps by combining IoT-based energy monitoring with POS-based expense tracking and predictive forecasting. This integrated approach provides a comprehensive solution for managing aquaculture operations more efficiently.

3 METHODOLOGY

3.1 Overview

This research follows a practical system development approach to design and implement an IoT-based cost analytics and forecasting solution for aquaculture. The methodology focuses on integrating real-time energy monitoring with structured expense tracking and predictive analysis. The system is designed to operate in small-scale farm environments where affordability, reliability, and ease of use are critical.

The overall process begins with data collection from sensors and manual inputs, followed by processing, storage, analysis, and visualization. The system then applies forecasting techniques to support future planning and decision-making.

3.2 System Architecture

The proposed system is based on a layered IoT architecture that connects physical devices, cloud services, and user applications. At the core of the system is the ESP32 microcontroller, which collects electrical data from sensors installed at the main power supply. This data is combined with manually entered expenses through a POS interface.

The collected data is transmitted to cloud services, where it is stored and processed. The backend system performs analytics and generates insights, which are then displayed through a web-based dashboard. This architecture ensures real-time monitoring, remote accessibility, and scalability.

3.3 Hardware Setup

The hardware configuration is designed to measure total energy consumption efficiently. The ESP32 microcontroller acts as the central processing unit, connected to current and voltage sensors. The ACS712 sensor measures the current flow, while the ZMPT101B sensor measures voltage levels.

These sensors are installed at the main power input of the farm, allowing the system to capture total electricity usage instead of monitoring individual devices. A relay-based interlock system is also included to protect equipment by preventing operation under unsafe voltage conditions.

3.4 Data Collection and Processing

The system collects two main types of data. The first type is electrical data, which includes voltage and current readings obtained from sensors. The second type is financial data, which includes operational expenses such as feed, labour, chemicals, and maintenance entered through the POS interface.

The ESP32 processes raw sensor data to calculate power and energy values. The basic relationship used for power calculation is:

$$P=V\times I$$

where power is derived from voltage and current. Energy consumption is then calculated over time and converted into cost using predefined tariff rates.

At the same time, expense data entered by the user is categorized and stored in the database. By combining these two data sources, the system creates a complete financial record of farm operations.

3.5 Cost Analysis and Key Performance Indicators

The system generates important financial indicators that help evaluate farm performance. Total cost is calculated by combining energy expenses with operational costs such as feed and labour. From this, the system derives metrics such as cost per production cycle and cost per fish.

The cost per fish is calculated as:

$$\text{Cost per Fish} = \text{Total Cost} / \text{Number of Fish}$$

In addition, the system analyzes how costs are distributed across different categories. This allows farmers to identify which components contribute most to overall expenses and take corrective actions if necessary.

3.6 Predictive Forecasting

To support future planning, the system uses a time-series forecasting approach based on the ARIMA model. This model analyzes historical energy consumption and cost data to predict future values.

Instead of relying on complex models that require large datasets, ARIMA is chosen because it performs well with limited data, which is typical in small-scale aquaculture farms. The system generates short-term forecasts along with confidence ranges, allowing users to estimate upcoming costs and plan budgets more effectively.

Forecast accuracy is evaluated using error measurements such as Mean Absolute Error (MAE), which compares predicted values with actual observations.

3.7 Communication and Data Flow

The system uses MQTT protocol for efficient communication between devices and the backend. The ESP32 publishes sensor data to the MQTT broker, which then forwards the data to the cloud server. The backend processes this data and updates the database in real time.

This communication model ensures low latency and reliable data transfer. It also supports remote monitoring, allowing users to access system data from anywhere using web applications.

3.8 Backend Processing and Storage

The backend system is responsible for managing data storage, processing, and analytics. A cloud-based database such as Supabase or SQL is used to store sensor readings, expense records, and calculated results.

The backend also handles API requests from the frontend, ensuring that users can retrieve data efficiently. Security mechanisms such as authentication and role-based access control are implemented to protect sensitive information.

3.9 Alert and Safety Mechanisms

The system includes an alert mechanism to notify users of abnormal conditions. Alerts are triggered when energy consumption exceeds expected levels, when costs deviate from the budget, or when voltage levels fall outside the safe operating range.

A voltage interlock mechanism is implemented to protect equipment. If the voltage exceeds or drops below the acceptable range, the system prevents pumps or motors from operating. This feature helps reduce equipment damage and ensures safe operation.

3.10 System Workflow

The operation of the system follows a continuous cycle. The ESP32 collects sensor data and processes it locally before sending it to the cloud. Users enter expense data through the POS interface, which is also stored in the database.

The backend combines these data sources to calculate costs and generate insights. Forecasting algorithms analyze historical data to predict future trends. Alerts are generated when necessary, and all information is displayed on the dashboard for user interaction.

3.11 Evaluation Plan

The performance of the system is evaluated based on several criteria. Accuracy of energy measurement is verified against a reference meter, with an acceptable error margin of around five percent. Forecasting performance is evaluated using error metrics, ensuring that predictions remain within a reasonable range.

System responsiveness is also measured, including the time taken to generate alerts and activate safety mechanisms. Reliability is assessed by testing continuous operation under real-world conditions.

3.12 Summary

This chapter presented the methodology used to develop the proposed system, including system architecture, hardware setup, data processing, cost analysis, and forecasting. The approach focuses on integrating IoT technology with financial analytics to provide a practical and efficient solution for aquaculture cost management.

4 SYSTEM DESIGN AND IMPLEMENTATION

4.1 Overview

This chapter presents the detailed design and implementation of the Budget Tracking, Cost Analytics, and Predictive Forecasting module developed as part of the IoT-enabled aquaculture system. The design focuses on integrating hardware components, software modules, and cloud infrastructure into a unified system that provides real-time monitoring and financial analysis.

The implementation combines embedded systems, IoT communication, backend services, and user interfaces to create a complete solution. Each component is designed to operate efficiently while maintaining simplicity and affordability, making the system suitable for small-scale aquaculture farms.

4.2 System Architecture Design

The overall system architecture is designed using a layered approach to ensure modularity and scalability. The architecture consists of the edge layer, communication layer, cloud layer, and application layer.

At the edge layer, the ESP32 microcontroller collects real-time electrical data from sensors connected to the main power supply. This data is processed locally to calculate power and energy values. The communication layer uses MQTT protocol to transmit data from the ESP32 to the cloud server.

The cloud layer handles data storage, processing, and analytics. A backend service manages incoming data, stores it in a database, and performs calculations such as cost estimation and forecasting. The application layer includes the web-based dashboard, which allows users to view system data, enter expenses, and monitor alerts.

This architecture ensures that data flows seamlessly from sensors to the user interface while maintaining real-time performance and reliability.

4.3 Hardware Design and Implementation

The hardware system is designed to measure electricity consumption at the main power source of the aquaculture farm. This approach reduces complexity compared to device-level monitoring while still providing accurate cost estimation.

The ESP32 microcontroller is used as the central processing unit due to its low power consumption, built-in Wi-Fi capability, and sufficient processing power for IoT applications. It interfaces with the ACS712 current sensor and the ZMPT101B voltage sensor to capture electrical parameters.

The ACS712 sensor measures the current flowing through the system, while the ZMPT101B sensor measures the voltage of the supply. These sensors provide analog signals that are read by the ESP32 and converted into digital values for processing.

A relay-based interlock mechanism is integrated into the system to ensure safety. This mechanism automatically disconnects the power supply to pumps and motors when the voltage falls outside the safe operating range. Additionally, indicator LEDs and a display module are used to provide real-time feedback on system status.

The hardware components are enclosed in a protective casing suitable for outdoor environments, ensuring durability and reliability under field conditions.

4.4 Software Design

The software component of the system is divided into multiple modules, each responsible for a specific functionality. These include the embedded firmware, backend services, and frontend application.

The embedded firmware running on the ESP32 is responsible for reading sensor data, performing basic calculations, and transmitting data to the cloud. It also handles control functions such as switching devices on and off based on user commands or safety conditions. The backend system is developed to process incoming data, perform analytics, and manage database operations. It provides APIs that allow the frontend application to retrieve and display data. The backend also implements logic for cost calculation, anomaly detection, and forecasting.

The frontend application is designed as a web-based dashboard that provides a user-friendly interface for monitoring and controlling the system. It allows users to view real-time energy consumption, enter expenses, and analyze financial data.

4.5 POS Module Design

The Point-of-Sale (POS) module is an essential part of the system, enabling users to record operational expenses in a structured manner. The design of this module focuses on simplicity and efficiency, allowing users to enter data quickly with minimal effort.

The POS interface includes input fields for expense category, quantity, unit cost, and additional notes. Categories such as feed, labour, chemicals, and maintenance are predefined, but users can also add custom categories if needed.

Each expense entry is associated with a specific production cycle and pond, allowing the system to organize data effectively. The module also includes validation mechanisms to ensure data accuracy and prevent incorrect entries.

All expense data is stored in the database and linked with sensor data to provide a comprehensive view of farm operations.

4.6 Database Design

The database is designed to store and manage all system data efficiently. It includes multiple tables to handle different types of information, such as sensor readings, expense records, user data, and forecast results.

The sensor data table stores time-stamped records of voltage, current, power, and energy values. The expense table stores details of all POS entries, including category, amount, and associated cycle.

A separate table is used to store calculated KPIs, such as total cost and cost per fish. Forecast results are also stored in a dedicated table for comparison with actual values.

Relationships between tables are defined to ensure data consistency and support efficient queries. The database is optimized for real-time updates and retrieval, enabling smooth operation of the dashboard.

4.7 Dashboard Design and Visualization

The dashboard is designed to provide clear and meaningful insights into farm operations. It presents data in the form of charts, graphs, and summary panels, allowing users to quickly understand system performance.

The main dashboard displays real-time energy consumption, total cost, and key performance indicators. Additional views provide detailed analysis of cost distribution across different categories and production cycles.

Forecasting results are displayed alongside actual data, enabling users to compare predictions with real outcomes. Interactive features allow users to filter data by date, category, or cycle.

The design emphasizes simplicity and readability, ensuring that even users with limited technical knowledge can easily interpret the information.

4.8 IoT Communication Implementation

The system uses MQTT protocol to enable communication between the ESP32 device and the cloud server. MQTT is chosen due to its lightweight nature and suitability for IoT applications. The ESP32 acts as a publisher, sending sensor data to specific topics on the MQTT broker. The backend system subscribes to these topics and processes incoming data in real time. This publish-subscribe model ensures efficient data transmission and reduces network overhead. It also allows the system to scale easily by adding more devices if needed.

4.9 Backend and Cloud Deployment

The backend system is deployed on a cloud-based Virtual Private Server (VPS) using Docker containers. This approach ensures flexibility, scalability, and ease of maintenance.

Services such as MQTT broker, database, and backend APIs are hosted within separate containers, allowing independent management of each component. The use of containerization also simplifies deployment and updates.

The cloud environment provides high availability and enables remote access to the system. Users can monitor and control the system from any location using the web dashboard.

4.10 Alert System Implementation

The alert system is implemented to notify users of critical conditions that require attention. Alerts are generated when predefined thresholds are exceeded, such as high energy consumption, budget overruns, or unsafe voltage levels.

Notifications are delivered through multiple channels, including the web dashboard, SMS, and email. The system ensures that alerts are timely and informative, enabling users to take immediate action.

To avoid repeated notifications, a cooldown mechanism is implemented, ensuring that alerts are not triggered multiple times for the same event within a short period.

4.11 Safety and Control Mechanisms

Safety is a key consideration in the system design. The voltage interlock mechanism ensures that electrical equipment operates within safe limits. If the voltage falls below or exceeds the acceptable range, the system automatically prevents the operation of pumps and motors.

In addition to automatic control, users can manually control devices through the dashboard. Scheduling features allow users to define operating times for equipment, improving efficiency and reducing unnecessary energy consumption.

4.12 Integration of System Components

The successful operation of the system depends on the integration of hardware, software, and communication components. Sensor data, expense records, and user inputs are combined to generate meaningful insights.

The system ensures synchronization between different components, allowing real-time updates and consistent data across all modules. This integration enables the system to function as a unified solution rather than a collection of independent parts.

4.13 Summary

This chapter described the design and implementation of the proposed system in detail. It covered the architecture, hardware setup, software modules, database design, and communication mechanisms.

The system is designed to provide a practical and efficient solution for cost management in aquaculture. By integrating IoT technology with financial analytics, the system offers real-time insights, predictive capabilities, and improved decision-making support.

5 TESTING AND EVALUATION

5.1 Overview

This chapter presents the testing procedures and evaluation methods used to validate the performance of the proposed Budget Tracking, Cost Analytics, and Predictive Forecasting system. The objective of testing is to ensure that the system operates accurately, reliably, and efficiently under real-world conditions.

The evaluation focuses on key aspects such as energy measurement accuracy, expense tracking reliability, forecasting performance, system responsiveness, and overall stability. A combination of functional testing and performance testing is used to verify that all system components meet the required objectives.

5.2 Testing Strategy

The testing approach is divided into three main stages: unit testing, integration testing, and system-level testing.

Unit testing is carried out to verify the functionality of individual components such as sensor readings, POS data entry, and backend processing. Each component is tested independently to ensure correct operation.

Integration testing is performed to evaluate how different modules interact with each other. This includes testing communication between the ESP32 device, MQTT broker, backend server, and dashboard interface.

System-level testing involves evaluating the complete system in a real or simulated environment. This stage ensures that the system performs as expected when all components are working together.

5.3 Functional Testing

Functional testing is conducted to verify that all features of the system operate correctly. The main functionalities tested include energy monitoring, expense entry, cost calculation, forecasting, and alert generation.

The system is tested by applying known inputs and verifying whether the outputs match expected results. For example, a fixed electrical load is applied to check whether the system calculates energy consumption accurately. Similarly, expense entries are added through the POS module to verify correct storage and calculation of total cost.

The results confirm that the system successfully performs all required functions, including real-time monitoring, cost analytics, and data visualization.

5.4 Energy Measurement Accuracy

One of the critical aspects of the system is the accuracy of energy measurement. To evaluate this, the system readings are compared with a reference energy meter under controlled conditions.

Different load levels are applied, and the system's measured values are recorded over time. The observed results show that the system maintains an accuracy within an acceptable error range of approximately $\pm 5\%$.

This level of accuracy is sufficient for practical applications in small-scale aquaculture, where precise cost estimation is more important than laboratory-level measurement precision.

5.5 POS Data Validation

The POS module is tested to ensure that expense entries are recorded accurately and consistently. Test cases include entering different types of expenses, modifying entries, and verifying stored data.

The system successfully validates user inputs, prevents incorrect data entry, and maintains structured records in the database. The ability to categorize expenses and link them to specific production cycles ensures reliable cost tracking.

This confirms that the POS module provides a dependable method for managing operational expenses.

5.6 Cost Calculation Verification

The cost calculation functionality is evaluated by comparing system-generated results with manually calculated values. Test scenarios include combining energy costs with operational expenses to determine total cost.

The results show that the system correctly calculates total production cost and derived metrics such as cost per fish. The calculations remain consistent across different test cases, confirming the reliability of the analytics module.

5.7 Forecasting Performance Evaluation

The forecasting model is tested using historical data collected from the system. The model generates predictions for future energy consumption and total cost, which are then compared with actual values.

The accuracy of predictions is evaluated using error measurements such as Mean Absolute Error (MAE). The results indicate that the forecasting model achieves an accuracy range of approximately 15% to 25%, which is acceptable for short-term predictions in small-scale environments.

Although forecasting accuracy may improve with more data over time, the current performance is sufficient to support budgeting and planning decisions.

5.8 Alert System Testing

The alert system is tested by simulating different abnormal conditions. These include sudden increases in energy consumption, exceeding budget thresholds, and voltage levels outside the safe range.

The system successfully detects these conditions and generates alerts within a short time. Notifications are delivered through the dashboard, and additional channels such as SMS and email are also verified.

The implementation of a cooldown mechanism ensures that repeated alerts are minimized, improving user experience.

5.9 Voltage Safety and Interlock Testing

The voltage protection mechanism is tested by artificially varying the supply voltage. When the voltage exceeds or drops below the defined safe range, the system correctly prevents the operation of connected devices.

The response time of the interlock system is observed to be within a few seconds, ensuring rapid protection of equipment. This confirms that the safety mechanism is effective in preventing damage to electrical components.

5.10 System Performance and Reliability

The overall performance of the system is evaluated under continuous operation. The system is tested for extended periods to ensure stability and reliability.

The ESP32 device operates continuously without failure, and the backend system maintains consistent data synchronization. The dashboard updates in real time, providing users with up-to-date information.

Network interruptions are also simulated to test system resilience. The system successfully stores data locally and synchronizes it once the connection is restored, ensuring no data loss.

5.11 Summary of Test Results

The results of testing demonstrate that the system meets the required performance criteria.

Energy measurement accuracy remains within acceptable limits, expense tracking is reliable, and cost calculations are consistent.

The forecasting model provides useful predictions, while the alert system ensures timely notifications. The overall system operates efficiently and reliably, making it suitable for practical deployment in aquaculture environments.

5.12 Summary

This chapter evaluated the performance of the proposed system through a series of tests. The results confirm that the system is capable of providing accurate, reliable, and real-time cost analytics and forecasting.

The testing process validates the effectiveness of the system in achieving its objectives and highlights its suitability for real-world applications.

6 RESULTS AND DISCUSSION

6.1 Overview

This chapter presents the results obtained from the implementation of the Budget Tracking, Cost Analytics, and Predictive Forecasting system. It also provides a detailed discussion of system outputs, performance, and overall effectiveness in addressing the identified problem.

The results are analyzed based on real-time data collected from the system, including energy consumption, expense records, cost calculations, and forecasting outputs. The objective is to evaluate how well the system supports decision-making and improves financial management in aquaculture.

6.2 Energy Monitoring Results

The system successfully captures real-time electrical data from the main power supply using the connected sensors. The dashboard displays voltage, current, power, and energy consumption values continuously.

The observed results show stable data acquisition under normal operating conditions. Variations in energy consumption are clearly visible during different time periods, especially when equipment such as pumps and aerators are switched on or off.

This real-time visibility allows users to identify high-energy usage periods and understand how operational activities affect electricity consumption. Compared to traditional monthly billing methods, the system provides much more detailed and actionable insights.

6.3 Expense Tracking Results

The POS module effectively records operational expenses in a structured format. Users can enter expenses related to feed, labour, chemicals, and maintenance, and the system stores this information in the database.

The results show that expense data is categorized correctly and linked to specific production cycles. This allows users to track spending patterns over time and compare different cycles. The structured nature of the POS system reduces errors associated with manual record keeping and ensures that all financial data is organized and accessible.

6.4 Cost Analytics Results

By combining energy data and expense records, the system generates meaningful cost analytics. The dashboard displays total cost, cost distribution across categories, and cost per fish.

The results indicate that feed cost and energy consumption are the major contributors to total production cost. This aligns with real-world observations in aquaculture, where feed and electricity are typically the largest expenses.

The system also identifies unusual cost patterns, such as sudden increases in energy usage or unexpected expenses. These insights help users take corrective actions to optimize resource usage and reduce waste.

6.5 Key Performance Indicators (KPIs)

The system calculates key performance indicators that provide a clear overview of farm performance. These include total cost per cycle and cost per fish.

The cost per fish metric is particularly useful for evaluating profitability. By comparing this value across different cycles, users can determine whether operational efficiency is improving or declining.

The KPI results demonstrate that the system provides valuable metrics that were not easily accessible using traditional methods.

6.6 Forecasting Results

The forecasting module generates predictions for future energy consumption and operational costs based on historical data. The results show that the model is able to capture general trends and provide reasonable estimates for short-term planning.

The predicted values follow the same pattern as actual data, although some variation is observed due to the limited dataset and external factors such as weather conditions and operational changes.

Despite these variations, the forecasts provide useful guidance for budgeting and resource planning. Users can anticipate future costs and adjust their operations accordingly.

6.7 Alert System Results

The alert system performs effectively in detecting abnormal conditions. When energy consumption exceeds expected levels or when voltage goes outside the safe range, the system generates notifications promptly.

The results show that alerts are delivered within a short time, allowing users to respond quickly. The system also avoids excessive notifications by using a cooldown mechanism. This feature improves system usability and ensures that alerts remain meaningful and actionable.

6.8 System Performance

The overall system performance is stable and reliable. The ESP32 device operates continuously without interruption, and data transmission through MQTT is consistent. The backend processes data efficiently, and the dashboard updates in real time. Users can access system data without noticeable delays, which is essential for monitoring and decision-making. The system also demonstrates resilience to temporary network issues by storing data locally and synchronizing it once the connection is restored.

6.9 Discussion

The results demonstrate that the proposed system successfully addresses the limitations of traditional aquaculture management methods. By integrating energy monitoring, expense tracking, and analytics, the system provides a comprehensive view of farm operations.

One of the key strengths of the system is its ability to combine different types of data into a single platform. This allows users to understand the relationship between resource usage and cost, which is not possible with separate systems.

The forecasting feature adds further value by enabling proactive planning. Although the accuracy of predictions can be improved with more data, the current implementation provides sufficient guidance for decision-making.

Another important aspect is the affordability and simplicity of the system. Unlike industrial solutions, the proposed system is designed specifically for small-scale farms, making it practical and accessible.

However, some limitations are observed. Forecasting accuracy depends on the availability of historical data, and sensor calibration is required to maintain measurement accuracy. These limitations can be addressed in future improvements.

7 SUMMARY

This chapter presented the results obtained from the system and discussed their significance. The findings confirm that the system provides accurate monitoring, reliable cost analytics, and useful forecasting capabilities.

The results demonstrate that the system can effectively support financial management in aquaculture, making it a valuable tool for improving efficiency and decision-making.

8 CONCLUSION AND FUTURE WORK

8.1 Overview

This chapter presents the overall conclusion of the research and summarizes the key outcomes of the developed Budget Tracking, Cost Analytics, and Predictive Forecasting system. It also discusses the limitations observed during the implementation and suggests possible improvements for future development.

8.2 Conclusion

This study focused on designing and developing a low-cost, IoT-based system to improve financial management in aquaculture. The proposed solution successfully integrates energy monitoring, expense tracking, cost analytics, and predictive forecasting into a single platform. The system addresses the major problem of fragmented cost management in small-scale aquaculture farms. By replacing manual record keeping and monthly billing analysis with real-time data collection and processing, the system provides a more accurate and detailed view of operational expenses.

The implementation demonstrates that energy consumption can be effectively monitored using sensors connected to an ESP32 microcontroller. When combined with structured expense data collected through the POS module, the system is able to calculate total production cost and key performance indicators such as cost per fish.

The forecasting component further enhances the system by enabling users to predict future costs based on historical data. This allows farmers to plan budgets in advance and make informed decisions regarding resource allocation.

The alert and safety mechanisms ensure that abnormal conditions are detected early, helping

to prevent equipment damage and reduce unnecessary expenses. The dashboard interface provides a simple and accessible way for users to interact with the system and understand key insights.

Overall, the system achieves its main objective of providing a practical and efficient solution for cost management in aquaculture. The results confirm that integrating IoT technology with financial analytics can significantly improve decision-making and operational efficiency.

8.3 Achievements of the Study

The study successfully achieved several important outcomes. A complete system was designed and implemented, combining hardware and software components into a unified solution. Real-time energy monitoring was established using sensors and embedded systems, enabling continuous tracking of electricity usage.

A structured POS module was developed to record operational expenses, ensuring accurate and organized data collection. The system also implemented cost analytics features that generate meaningful insights such as total cost, cost distribution, and cost per fish.

The forecasting module was able to predict future energy consumption and costs using time-series analysis, providing valuable support for budgeting. In addition, the system includes alert mechanisms and safety features that enhance reliability and usability.

These achievements demonstrate the effectiveness of the proposed approach and its suitability for real-world applications.

8.4 Limitations of the Study

Although the system performs effectively, several limitations were identified during the implementation. The accuracy of energy measurement depends on proper calibration of sensors, and small deviations may occur under certain conditions.

The forecasting model is based on a limited dataset, which affects prediction accuracy. As more data becomes available over time, the performance of the model is expected to improve. External factors such as weather conditions and changes in farm operations can also influence results.

Another limitation is that the system is designed primarily for small-scale aquaculture farms. It may require modifications to handle large-scale industrial operations with multiple ponds and complex infrastructure.

Additionally, the current implementation focuses mainly on web-based access, and a dedicated mobile application could further improve usability.

8.5 Future Work

Several enhancements can be made to improve the system in future developments. One important area is the improvement of forecasting accuracy by incorporating advanced machine learning models and larger datasets.

The system can also be extended to support multi-pond management, allowing users to monitor and compare multiple production units simultaneously. Integration with additional sensors and automation features can further enhance system capabilities.

Developing a mobile application would improve accessibility and user experience, enabling farmers to monitor their operations more conveniently. Cloud-based analytics can also be expanded to include advanced reporting and benchmarking features.

Another potential improvement is the integration of cost optimization algorithms that provide recommendations for reducing expenses and improving efficiency.

8.6 Final Remarks

This research demonstrates that the integration of IoT technology with cost analytics and forecasting can significantly improve aquaculture management. The proposed system provides a practical, affordable, and scalable solution that addresses real challenges faced by small-scale farmers.

By enabling real-time monitoring, structured expense tracking, and predictive analysis, the system supports better decision-making and promotes more efficient use of resources. The study highlights the potential of smart technologies in transforming traditional aquaculture practices into data-driven systems.

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